A certified multilevel lower bound for the Galerkin prediction error of a log-singular Toeplitz form with a prime-power comb

Stefan Hamann

October 18, 2026

Abstract

For a reflection-odd compression A of a finite Toeplitz form and a data vector b, the prediction defect $\varepsilon = 1 - b^{\mathsf{T}} A^{-1} b$ is the Galerkin error of a piecewise-constant approximation to a fixed function, and bounding it from *below* is the hard direction. Our result is a single inequality: on a nested ladder of resolutions the increment of the saturation telescope admits a certified lower bound by one explicit test vector, in closed form. The mechanism is a change of direction — the increment is a residual measured in the *inverse* energy norm, hence a *maximum* over test vectors rather than a minimum, so one test vector already suffices and the coarse-level block enters only through its sign, never through its smallest eigenvalue, its condition number, or a strengthened Cauchy–Schwarz constant. The proof combines Schur complement monotonicity, Parseval against a certified majorant, and Löwner's antitony of the inverse; the nesting the ladder needs is proved as an identity for any lag family generated by a fixed kernel. Two features of the statement are recorded up front rather than in a remark. First, a scope: the numerator of the certified rung is evaluated at the coarse Galerkin solution carried along the ladder, which is the information an a posteriori estimator has, so the bound is a bound *given* the ladder. Second, a measured negative result: the solution-free variant of the same rung fails, not by a small margin but by cancellation — the coupling term annihilates the explicit high-pass part of the data instead of perturbing it, and a norm-only Cauchy–Schwarz estimate of it returns exactly zero of the rung on every rung of the tested ladder. As an application we compose the certified rungs with a margin-free bordering criterion into a chain that transports semidefiniteness along a nested family of windows, and we report a verified instance of that chain in which every load-bearing step is a completed Cholesky factorisation above an explicitly declared floating-point floor: 430 such certificates over 52 windows and 88 rungs. The chain is verified window by window and is not claimed uniformly in the window index. The example class that motivates the construction has a symbol built from a logarithmic profile minus an explicit cosine comb at the frequencies $\log n$, n a prime power; the arithmetic enters only as the source of the comb positions and only through two classical gap inequalities.

1 Introduction

1.1 The problem, and why the direction matters

Let $A \in \mathbb{R}^{h\times h}$ be symmetric positive definite, let $b \in \mathbb{R}^h$, and set

$$\varepsilon \;=\; 1 - b^{\mathsf{T}} A^{-1} b, \qquad Q \;=\; A - b b^{\mathsf{T}}. \tag{1}$$

The scalar ε is the value of the quadratic functional $J(v) = 1 - 2b^{\mathsf{T}} v + v^{\mathsf{T}} A v$ at its minimiser $u = A^{-1} b$; when A is a Galerkin matrix on a finite-dimensional space V and b the load vector of a fixed right-hand side, ε is exactly the Galerkin error of the best approximation in V, measured in the energy of the ambient form. When A is a Toeplitz matrix and b the first coordinate

vector, ε is the classical one-step prediction error of Szegő, Levinson and Durbin [27, 16, 21, 12], whence the name we use throughout. Finally, and this is the elementary fact the whole paper turns on, by Albert's criterion applied to the two Schur readings of the bordered matrix $\left[\begin{smallmatrix} A & b \\ b^{\mathsf{T}} & 1 \end{smallmatrix}\right]$ one has, for $A \succ 0$,

$$\varepsilon \geq 0 \quad \Longleftrightarrow \quad \begin{bmatrix} A & b \\ b^{\mathsf{T}} & 1 \end{bmatrix} \succeq 0 \quad \Longleftrightarrow \quad Q \succeq 0. \tag{2}$$

So a *lower* bound on ε is a statement about the semidefiniteness of a rank-one-perturbed form, and vice versa.

Upper bounds for ε are the classical direction. Céa's lemma, duality, and the Bramble–Hilbert machinery all estimate an approximation error from above [8]. Lower bounds are the hard direction, and the standard device for obtaining them — the *saturation assumption*, which asserts that refining the space reduces the error by a factor bounded away from one — is known to be delicate: Bank and Smith made it the basis of hierarchical a posteriori estimators [3], Bornemann, Erdmann and Kornhuber isolated the constants it consumes [5], and Dörfler and Nochetto showed that it holds under a smallness condition on the data oscillation and can fail otherwise [10]. In the multilevel literature the same difficulty appears as the strengthened Cauchy–Schwarz constant $\gamma^2 < 1$ of hierarchical splittings [32, 6, 2, 7]: it is a *worst-case* quantity, an upper bound on a coupling, and any bound that has to supply it inherits its worst case.

There is a second, more specific obstruction in the Toeplitz setting, and it is what makes the example class of Section 2.4 genuinely hard rather than merely technical. If the symbol σ of the form is nonnegative, the Toeplitz matrix $T_M(\sigma)$ is positive semidefinite, and conversely a pointwise minorant $\ell \leq \sigma$ transfers to an operator minorant $T_M(\ell) \preceq T_M(\sigma)$; this is Parseval together with the Fejér–Riesz representation of nonnegative trigonometric polynomials [13, 26], and it is the standard route from a symbol to a spectral bound [19, 16]. The route is one-sided in a way that matters here. For the forms we care about the smallest eigenvalue is not produced by a small value of the symbol at some frequency; it is produced by *cancellation inside the quadratic form*, along a direction that is smooth and lives in the coarse part of the space. No pointwise minorant of the symbol can see such a direction, because a pointwise minorant is by construction blind to which vector realises the minimum. Section 2.4 records the measured form of this statement for the example class; the theorem of Section 4 is designed so that it never needs a symbol minorant at all.

1.2 Contribution

The result of the paper is (C1). The rest is what (C1) is good for, and what it costs.

(C1) **The result: a direction reversal that makes a certified rung possible** (Theorem 4.4). On a nested ladder, the increment $\delta_l = \varepsilon_l - \varepsilon_{l+1}$ of the saturation telescope is the squared *inverse*-energy norm of the level residual, hence a maximum over test vectors. A single test vector therefore supplies a valid lower bound, and the denominator of that bound wants the relevant Schur complement from *above*. Schur complement monotonicity [17] supplies the upper bound from the mere semidefiniteness of the coarse block, Parseval supplies an upper bound by a certified majorant of the symbol, and Löwner's antitony of the matrix inverse [22] closes the chain. No coarse inverse, no $\lambda_{\min}$ of a coarse block, and no strengthened Cauchy–Schwarz constant enters. The optimal test vector for the majorant is available in closed form, and the entire loss relative to δ_l is the loss in the majorant. The nesting the theorem needs is not assumed: Lemma 2.7 proves it for every lag family generated by a fixed kernel, and Lemma 2.9 exhibits the kernel for the example class.

(C2) **The scope of the result, stated as such.** The numerator of the certified rung is evaluated at the coarse Galerkin solution carried along the ladder (Convention 4.7),

which is exactly the information an a posteriori estimator has and no more: it involves neither the fine solution nor ε itself. Remark 4.8 spells this out, and Proposition 6.1 measures the distance to the stronger, a priori object: the solution-free variant of the same rung fails through cancellation, which we report as a negative result on a specific ladder because it is what tells a reader that the convention was not an oversight.

(C3) **An application: margin-free composition into a chain** (Corollary 5.6), together with a verified instance of it. Two exact identities (Lemmas 5.2 and 5.3) show that on a nested family of windows at one common resolution the trailing block of the next window's form *is* the current window's form. Albert's criterion [1] with Douglas' range condition [11] then transports semidefiniteness from one window to the next using only the *sign* of that block — never the size of ε — and the residual condition is a Schur complement of the small dimension of the window increment. The hypotheses of that corollary are verified window by window on the instance of Section 7; a version uniform in the window index is not claimed, and item (L2) of Section 8 says precisely what is missing for one.

The instance is produced in Section 7 by a single script, runtime about ten seconds: 430 completed Cholesky certificates and the identity residuals, each stated relative to a declared floating-point floor in the sense of Wilkinson and Higham [31, 18]. We are careful throughout to distinguish what is proved (the identities and Theorem 4.4) from what is certified on one finite instance (the hypotheses of Corollary 5.6) from what is merely measured or fitted.

1.3 Notation

$T_M(c) = (c_{|r-s|})_{r,s=0}^{M-1}$ is the symmetric Toeplitz matrix of a real sequence $c = (c_m)_{m\geq 0}$. For real integrable even g on $[-\pi,\pi]$ with Fourier coefficients $\widehat{g}_m$ we write $T_M(g) := T_M(\widehat{g})$; no confusion arises because a symbol enters only through its first M coefficients. $X \succeq 0$ and $X \succ 0$ denote positive semidefiniteness and definiteness, $X \preceq Y$ means $Y - X \succeq 0$, X^+ is the Moore–Penrose pseudoinverse, and $\|v\|_X^2 = v^{\mathsf{T}} X v$ for $X \succeq 0$. Finally $(t)_+ = \max(t, 0)$ and $\Delta_D(s) = (1 - |s|/D)_+$ is the unit-height hat of half-width D.

2 Setting

2.1 The odd window form and its defect

Fix a cell width $D > 0$ and an even $M \in 2\mathbb{N}$, and put $\alpha = MD/2$ and $h = M/2$. The grid is the partition of $(-\alpha, \alpha)$ into M cells of width D with midpoints

$$\bar{x}_r \;=\; -\alpha + (r + \tfrac{1}{2})D, \qquad r = 0, \dots, M-1, \tag{3}$$

and χ_r denotes the L^2-normalised indicator of cell r, i.e. $\chi_r = D^{-1/2}\mathbf{1}_{[\bar{x}_r - D/2,\, \bar{x}_r + D/2]}$. Let $J \in \mathbb{R}^{M\times M}$ be the reflection $(Jv)_r = v_{M-1-r}$ and let

$$B \in \mathbb{R}^{M\times h}, \qquad Be_r = \tfrac{1}{\sqrt{2}}\,(e_r - e_{M-1-r}), \qquad r = 0, \dots, h-1, \tag{4}$$

be the isometry onto the reflection-odd sector, so $B^{\mathsf{T}}B = I_h$ and $JB = -B$.

Definition 2.1 (window form and pole vector)**.** Given a real sequence $c = (c_m)_{m=0}^{M-1}$, the *odd window form* and the *pole vector* at (D, M) are

$$A \;=\; B^{\mathsf{T}}\, T_M(c)\, B \;\in\; \mathbb{R}^{h\times h}, \qquad A_{rs} = c_{|r-s|} - c_{M-1-r-s}, \tag{5}$$

$$b \;=\; (b_r)_{r=0}^{h-1}, \qquad b_r = \frac{8}{\sqrt{D}}\,\sinh\Big(\frac{D}{4}\Big)\sinh\Big(\frac{\bar{x}_r}{2}\Big). \tag{6}$$

The associated *defect* and *bordered form* are ε and Q of (1).

The particular constant in (6) is not a normalisation choice; it is the reason the whole construction is a Galerkin problem on nested spaces.

Lemma 2.2 (b is the coefficient vector of a fixed function)**.** *Let $f(x) = 2\sinh(x/2)$. Then $b_r = \langle f, \chi_r\rangle_{L^2(\mathbb{R})}$ for $r = 0, \dots, h-1$, for every $D > 0$ and every M.*

Proof. $\langle f, \chi_r\rangle = D^{-1/2}\int_{\bar{x}_r - D/2}^{\bar{x}_r + D/2} 2\sinh(x/2)\,dx = D^{-1/2}\big[4\cosh(x/2)\big]_{\bar{x}_r - D/2}^{\bar{x}_r + D/2}$, and $\cosh(a+d) - \cosh(a-d) = 2\sinh a \sinh d$ with $a = \bar{x}_r/2$, $d = D/4$ gives $8D^{-1/2}\sinh(D/4)\sinh(\bar{x}_r/2)$. □

Remark 2.3. Write $V_{D,M} \subset L^2(-\alpha, \alpha)$ for the space of odd piecewise-constant functions on the grid (3); $\{B\chi\}$ is an orthonormal basis of it, and $\dim V_{D,M} = h$. If $a(\cdot,\cdot)$ denotes the bilinear form whose matrix in that basis is A, then by Lemma 2.2

$$\varepsilon \;=\; \min_{v \in V_{D,M}} \Big(1 - 2\langle f, v\rangle + a(v,v)\Big), \tag{7}$$

so ε is the Galerkin defect of the fixed function f in the space $V_{D,M}$, and its minimiser is the Galerkin solution $u = A^{-1}b$. Nothing in (7) refers to c except through a.

2.2 The ladder and exact nesting

We refine at fixed α. Let $L \geq 1$ and

$$D_l = 2^{-l}D_0, \qquad M_l = 2^l M_0, \qquad h_l = M_l/2, \qquad l = 0, \dots, L, \tag{8}$$

so $\alpha = M_l D_l/2$ is the same at every level and $V_{D_0,M_0} \subset V_{D_1,M_1} \subset \cdots \subset V_{D_L,M_L}$. Write $A^{(l)}, b^{(l)}, \varepsilon_l, u^{(l)}$ for the objects of Definition 2.1 at level l. Let $P_l : \mathbb{R}^{h_l} \to \mathbb{R}^{h_{l+1}}$ be the matrix of the inclusion $V_{D_l,M_l} \hookrightarrow V_{D_{l+1},M_{l+1}}$ in the orthonormal bases, i.e. the L^2-normalised piecewise-constant prolongation: each coarse cell splits into two fine cells and

$$(P_l x)_{2i} = (P_l x)_{2i+1} = \tfrac{1}{\sqrt{2}}\, x_i. \tag{9}$$

We suppress the level index on P when it is clear. Let $Z_l : \mathbb{R}^{h_l} \to \mathbb{R}^{h_{l+1}}$ be the complementary Haar injection,

$$(Z_l x)_{2i} = \tfrac{1}{\sqrt{2}}\, x_i, \qquad (Z_l x)_{2i+1} = -\tfrac{1}{\sqrt{2}}\, x_i, \tag{10}$$

so that

$$P^{\mathsf{T}}P = Z^{\mathsf{T}}Z = I_{h_l}, \qquad P^{\mathsf{T}}Z = 0, \qquad \operatorname{ran} P \oplus \operatorname{ran} Z = \mathbb{R}^{h_{l+1}}. \tag{11}$$

The pair (P, Z) is a two-level orthogonal splitting of the fine space into its piecewise-constant (coarse) and oscillatory (high-pass) parts.

The property we need of the ladder is that it is *one* bilinear form on nested spaces:

$$P^{\mathsf{T}}A^{(l+1)}P = A^{(l)}, \qquad P^{\mathsf{T}}b^{(l+1)} = b^{(l)} \qquad (l < L). \tag{12}$$

Both halves of (12) are identities, and we prove them. The second is unconditional.

Lemma 2.4 (nesting of the pole vector)**.** *$P^{\mathsf{T}}b^{(l+1)} = b^{(l)}$ holds for every D_l and every M_l.*

Proof. Immediate from Lemma 2.2, since P is the matrix of an inclusion of spaces in orthonormal bases and f is fixed. Directly: with $d = D_l/2$ the two fine midpoints inside the coarse cell centred at $\bar{x}$ are $\bar{x} \mp d/2$, so

$$(P^{\mathsf{T}}b^{(l+1)})_i = \frac{1}{\sqrt{2}}\cdot\frac{8}{\sqrt{d}}\sinh\frac{d}{4}\Big[\sinh\frac{\bar{x} - d/2}{2} + \sinh\frac{\bar{x} + d/2}{2}\Big] = \frac{1}{\sqrt{2}}\cdot\frac{8}{\sqrt{d}}\sinh\frac{d}{4}\cdot 2\cosh\frac{d}{4}\,\sinh\frac{\bar{x}}{2},$$

which equals $8\,(2d)^{-1/2}\sinh(d/2)\sinh(\bar{x}/2) = b_i^{(l)}$ because $D_l = 2d$. □

The first half holds for every lag family generated by a fixed kernel, in the following sense.

Definition 2.5 (consistent lag family)**.** Let K be an even distribution on $\mathbb{R}$ for which $K * \Delta_D$ is a well-defined continuous function for every $D > 0$. The family $\{c^{(l)}\}_{l=0}^{L}$ is *generated by* K if

$$c_m^{(l)} \;=\; (K * \Delta_{D_l})(mD_l), \qquad 0 \le m \le M_l - 1, \quad 0 \le l \le L, \tag{13}$$

with the *same* K at every level. We then call the family *consistent*.

Condition (13) is not an extra hypothesis dressed up as a definition; it is what the phrase "the Toeplitz matrix of a kernel" means. Indeed if $T_{rs} = \iint K(x-y)\chi_r(x)\chi_s(y)\,dx\,dy$ is the Galerkin matrix of convolution by K in the cell basis, then substituting $t = y - x$ and using that the convolution of two indicators of width D shifted by mD is $D\,\Delta_D(\cdot - mD)$ gives $T_{rs} = (K * \Delta_D)((r-s)D)$, i.e. exactly (13). The even extension $c_{-m}^{(l)} := c_m^{(l)}$ is automatic, because K even implies $K * \Delta_D$ even.

Lemma 2.6 (hat refinement)**.** *For every $d > 0$ and every $u \in \mathbb{R}$,*

$$\Delta_d(u) + \tfrac{1}{2}\Delta_d(u-d) + \tfrac{1}{2}\Delta_d(u+d) \;=\; \Delta_{2d}(u). \tag{14}$$

Proof. Both sides are continuous, piecewise linear, supported in $[-2d, 2d]$, and their breakpoints are contained in $\{-2d, -d, 0, d, 2d\}$: the left-hand side because each of its three terms is piecewise linear with breakpoints in that set, the right-hand side because Δ_{2d} has breakpoints $\{-2d, 0, 2d\}$. Two continuous piecewise-linear functions with breakpoints in a common finite set agree everywhere as soon as they agree at every point of that set. At $u = 0$ the left-hand side is $1 + \frac{1}{2}\cdot 0 + \frac{1}{2}\cdot 0 = 1 = \Delta_{2d}(0)$; at $u = \pm d$ it is $0 + \frac{1}{2}\cdot 1 + 0 = \frac{1}{2} = \Delta_{2d}(\pm d)$; at $u = \pm 2d$ both sides vanish. □

Lemma 2.7 (exact nesting)**.** *Let $\{c^{(l)}\}$ be consistent in the sense of Definition 2.5. Then* (12) *holds for every $l < L$.*

Proof. Step 1: reduction to the full Toeplitz level. Let $P_{\text{full}} : \mathbb{R}^{M_l} \to \mathbb{R}^{M_{l+1}}$ be the prolongation (9) on the unreduced grid. Then $B_{M_{l+1}} P = P_{\text{full}} B_{M_l}$. Indeed, using $M_{l+1} = 2M_l$,

$$\begin{aligned} P_{\text{full}} B_{M_l} e_r &= \tfrac{1}{\sqrt{2}} P_{\text{full}}\big(e_r - e_{M_l - 1 - r}\big) \\ &= \tfrac{1}{2}\Big[\big(e_{2r} - e_{M_{l+1}-1-2r}\big) + \big(e_{2r+1} - e_{M_{l+1}-1-(2r+1)}\big)\Big] = B_{M_{l+1}} \tfrac{e_{2r} + e_{2r+1}}{\sqrt{2}}, \end{aligned}$$

because $2(M_l - 1 - r) = M_{l+1} - 1 - (2r+1)$ and $2(M_l - 1 - r) + 1 = M_{l+1} - 1 - 2r$; and the last expression is $B_{M_{l+1}} P e_r$ by (9). Consequently

$$P^{\mathsf{T}} A^{(l+1)} P = \big(B_{M_{l+1}} P\big)^{\mathsf{T}} T_{M_{l+1}}(c^{(l+1)}) \big(B_{M_{l+1}} P\big) = B_{M_l}^{\mathsf{T}} \Big(P_{\text{full}}^{\mathsf{T}} T_{M_{l+1}}(c^{(l+1)}) P_{\text{full}}\Big) B_{M_l}, \tag{15}$$

so it suffices to prove $P_{\text{full}}^{\mathsf{T}} T_{M_{l+1}}(c^{(l+1)}) P_{\text{full}} = T_{M_l}(c^{(l)})$.

Step 2: the coarsening is a three-term average. Writing $c' = c^{(l+1)}$ and using that the only nonzero entries of P_{full} in column i are the two rows $2i, 2i+1$, each equal to $2^{-1/2}$,

$$\big(P_{\text{full}}^{\mathsf{T}} T_{M_{l+1}}(c') P_{\text{full}}\big)_{ik} = \tfrac{1}{2} \sum_{a \in \{2i, 2i+1\}} \; \sum_{b \in \{2k, 2k+1\}} c'_{|a-b|} = c'_{2|m|} + \tfrac{1}{2}\big(c'_{|2m-1|} + c'_{|2m+1|}\big), \qquad m = i - k,$$

since the four terms are $c'_{|2m|}$, $c'_{|2m-1|}$, $c'_{|2m+1|}$ and $c'_{|2m|}$. The largest index occurring is $2(M_l - 1) + 1 = M_{l+1} - 1$, so every entry used is available. The matrix on the left is therefore Toeplitz, and $P_{\text{full}}^{\mathsf{T}} T_{M_{l+1}}(c') P_{\text{full}} = T_{M_l}(\tilde{c}\,)$ with

$$\tilde{c}_n = c'_{2n} + \tfrac{1}{2}\big(c'_{2n-1} + c'_{2n+1}\big), \qquad 0 \le n \le M_l - 1, \tag{16}$$

under the even extension $c'_{-1} := c'_1$.

Step 3: consistency turns (16) *into* $\tilde{c} = c^{(l)}$. Put $d = D_{l+1} = D_l/2$. By (13) and $(K * g)(x + a) = (K * g(\cdot + a))(x)$,

$$\tilde{c}_n = \big(K * \Delta_d\big)(2nd) + \tfrac{1}{2}\big(K * \Delta_d\big)\big(2nd - d\big) + \tfrac{1}{2}\big(K * \Delta_d\big)\big(2nd + d\big) = \big(K * G_d\big)(2nd),$$

where $G_d(u) = \Delta_d(u) + \frac{1}{2}\Delta_d(u + d) + \frac{1}{2}\Delta_d(u - d)$. By Lemma 2.6, $G_d = \Delta_{2d} = \Delta_{D_l}$, whence $\tilde{c}_n = (K * \Delta_{D_l})(nD_l) = c_n^{(l)}$. Together with (15) this is the first half of (12); the second half is Lemma 2.4. □

The example class of Section 2.4 is consistent, with an explicit K (Lemma 2.9), so (12) is an identity there and not a hypothesis. What Section 7 reports for it is a floating-point statement about two different orders of summation for the same exact quantity: the matrix $P^{\mathsf{T}} A^{(l+1)} P$, assembled at the fine level and then coarsened, agrees with $A^{(l)}$, assembled directly at the coarse level, to relative $3.0 \cdot 10^{-14}$.

2.3 The two-level split of the fine form

Fix a level l and abbreviate $A_f = A^{(l+1)}$, $V = P$, $W = Z$. Setting

$$A_c = V^{\mathsf{T}} A_f V, \qquad A_z = W^{\mathsf{T}} A_f W, \qquad B_x = V^{\mathsf{T}} A_f W, \tag{17}$$

the orthogonal splitting (11) puts the fine form in block shape $\left[\begin{smallmatrix} A_c & B_x \\ B_x^{\mathsf{T}} & A_z \end{smallmatrix}\right]$, and the nesting (12) identifies $A_c = A^{(l)}$: *the coarse block of a rung is the window form of the level below it.* The Schur complement onto the oscillatory space is

$$S \;=\; A_z - B_x^{\mathsf{T}} A_c^{-1} B_x. \tag{18}$$

If $A_f \succ 0$ then $A_c = V^{\mathsf{T}} A_f V \succ 0$ for the isometry V, so (18) is well defined, and $S \succ 0$.

2.4 The example class

The construction above is symbol-agnostic; Theorem 4.4 and Corollary 5.6 use no property of c beyond what is stated in their hypotheses. The class we verify on, and which motivates the whole question, is the following.

Definition 2.8 (the comb class)**.** Let $a : \mathbb{R} \to \mathbb{R}$ be the even kernel

$$a(s) = -(\gamma + \log \pi)\, \Delta_D(s) - \Delta_D(s) \log\!\big(1 - e^{-2W}\big) + 2 \int_0^W \frac{\Delta_D(s) e^{-2w} - S_D(s, w)\, e^{-w/2}}{1 - e^{-2w}}\, dw, \tag{19}$$

where $W = |s| + D$, $S_D(s, w) = \frac{1}{2}\big[\Delta_D(s - w) + \Delta_D(s + w)\big]$, and γ is Euler's constant. Let $\mathcal{U} = \{(u_j, \mu_j)\}_j$ be a finite set of *comb nodes* $u_j > 0$ with weights $\mu_j > 0$, and put

$$c_m \;=\; a(mD) \;-\; \sum_{j:\, u_j \le 2\alpha} \mu_j\, S_D(mD, u_j), \qquad m = 0, \dots, M - 1. \tag{20}$$

The *prime-power instance* is $u_j = \log n_j$ over the prime powers $n_j = p^\nu$ in increasing order, with $\mu_j = 2\Lambda(n_j)/\sqrt{n_j}$ and Λ the von Mangoldt weight $\Lambda(p^\nu) = \log p$.

The form (19) is written the way it is computed: the upper limit $W = |s| + D$ truncates the integral at the last point where the second numerator can be nonzero, and the $\log(1 - e^{-2W})$ term is the exact remainder of the tail. Undoing that bookkeeping identifies the fixed kernel behind the class.

Lemma 2.9 (the comb class is consistent)**.** *Fix α and a node set $\mathcal{U}$, and let $c^{(l)}$ be given by* (20) *at (D_l, M_l) along the ladder* (8)*. Then $\{c^{(l)}\}$ is consistent in the sense of Definition 2.5, generated by the even distribution*

$$K = \underbrace{-(\gamma + \log\pi)\,\delta_0 + 2\int_0^\infty \frac{e^{-2w}\delta_0 - e^{-w/2}\,\frac{1}{2}(\delta_w + \delta_{-w})}{1 - e^{-2w}}\,dw}_{K_{\mathrm{arch}}} \;-\; \sum_{j:\,u_j \le 2\alpha} \mu_j\,\tfrac{1}{2}(\delta_{u_j} + \delta_{-u_j}), \quad (21)$$

which does not depend on l. Consequently Lemma 2.7 applies and (12) *holds identically on the ladder.*

Proof. Both parts are pairings of a fixed even distribution against the hat, since $(\delta_v * \Delta_D)(s) = \Delta_D(s - v)$ and hence $(\frac{1}{2}(\delta_v + \delta_{-v}) * \Delta_D)(s) = S_D(s, v)$.

The comb part. The node set in (20) is $\{u_j \le 2\alpha\}$, and $\alpha = M_l D_l/2$ is the *same* at every level of the ladder (8); so the sum in (20) runs over one and the same finite set for all l, and it equals $(\sum_j \mu_j \frac{1}{2}(\delta_{u_j} + \delta_{-u_j}) * \Delta_{D_l})(mD_l)$ by the displayed identity.

The archimedean part. Fix s and let $w > W = |s| + D$. Then $\Delta_D(s - w) = 0$, because $w - s \ge w - |s| > D$, and $\Delta_D(s + w) = 0$, because $s + w \ge w - |s| > D$; hence $S_D(s, w) = 0$ and the integrand of (19) reduces to $\Delta_D(s)e^{-2w}/(1 - e^{-2w})$ there. Since $\frac{d}{dw}\log(1 - e^{-2w}) = 2e^{-2w}/(1 - e^{-2w})$,

$$2\int_W^\infty \frac{\Delta_D(s)e^{-2w}}{1 - e^{-2w}}\,dw = \Delta_D(s)\Big[\log(1 - e^{-2w})\Big]_W^\infty = -\Delta_D(s)\log(1 - e^{-2W}),$$

which is exactly the second term of (19). Therefore

$$a(s) = -(\gamma + \log\pi)\Delta_D(s) + 2\int_0^\infty \frac{\Delta_D(s)e^{-2w} - S_D(s, w)\,e^{-w/2}}{1 - e^{-2w}}\,dw, \quad (22)$$

an integral that converges absolutely at $w = 0$ because the numerator vanishes there ($S_D(s, 0) = \Delta_D(s)$) and is $O(w)$, while the denominator is $\sim 2w$; and at $w = \infty$ because of $e^{-w/2}$. Reading (22) through the displayed identity gives $a = K_{\mathrm{arch}} * \Delta_D$ with K_{arch} as in (21), independent of D. Adding the two parts and sampling at $s = mD_l$ gives (13). □

Remark 2.10. Lemma 2.9 is the reason this paper carries no nesting assumption. It also isolates what the ladder (8) is for: refining at *fixed* α keeps the admitted node set constant, and a family whose node set changed with the level would not be consistent.

Three comments on the class, all of them limitations of scope rather than claims.

(i) The label "log-singular". The integrand of (19) carries the factor $(1 - e^{-2w})^{-1} = \sum_{n\ge 0} e^{-2nw}$, whose expansion is the digamma series, and the constant is $-\gamma - \log\pi$. The kernel is therefore a logarithmic (digamma) profile smoothed against the hat Δ_D, and the label in the title refers to that profile. The integrand vanishes at $w = 0$ because $S_D(s, 0) = \Delta_D(s)$, so a is finite for every s; the symbol $\sigma^{(M)}(\theta) = \sum_{|m|\le M-1} c_{|m|}e^{im\theta}$ is a trigonometric polynomial for each fixed M, and no asymptotic property of the family is used anywhere below. The classical regularity theory for symbols of this type is Widom's and Fisher–Hartwig's [30, 14, 4]; we name it as the address of the questions we do *not* answer, not as an ingredient.

(ii) What the comb does. By (20) the symbol of c is a logarithmic profile minus an explicit cosine comb: each node contributes a term proportional to $\cos(u_j\theta/D)$ modulated by the hat's Fourier transform. This is the structural reason a pointwise symbol minorant is useless here. The comb dips are numerous and narrow — narrow enough that the large-sieve budget one would naturally spend on them [23] is exhausted before the window is covered — the infimum of the symbol over a window is driven to zero by them, and the near-null direction of the resulting form is nevertheless *smooth*: it lives in the coarse part $\operatorname{ran} V$ and its Rayleigh quotient is small

because of cancellation inside the form, not because the symbol is small along it. The measured statement is in Section 7.

(iii) What the arithmetic is used for. Only the positions $u_j = \log n_j$, and only through two classical inequalities on the consecutive log-gaps $g_k = u_{k+1} - u_k$, namely $g_k \le \log 2$ (Bertrand's postulate in Chebyshev's form [9]) and $g_k \ge \log(1 + 1/n_k)$. Both are used solely to check that a resolution is admissible in the sense of Definition 5.1; no unproved property of the sequence (u_j) enters the construction, and no arithmetic statement is claimed.

3 The saturation telescope

The identities of this section are classical; we record them in the form we use.

Lemma 3.1 (saturation identity and telescope)**.** *Assume $A^{(l)} \succ 0$ for all l and the nesting* (12)*. Put*

$$\delta_l \;=\; \|u^{(l+1)} - Pu^{(l)}\|^2_{A^{(l+1)}} \;\ge\; 0. \tag{23}$$

Then for every $l < L$

$$\varepsilon_l \;=\; \varepsilon_{l+1} + \delta_l, \qquad \textit{hence} \qquad \varepsilon_0 \;=\; \varepsilon_L + \sum_{l=0}^{L-1} \delta_l \quad \textit{(exactly).} \tag{24}$$

Moreover, with the level residual $r_l = b^{(l+1)} - A^{(l+1)} P u^{(l)}$,

$$V^{\mathsf T} r_l = 0, \qquad \delta_l = \|r_l\|^2_{(A^{(l+1)})^{-1}}. \tag{25}$$

Proof. Let $J_{l+1}(v) = 1 - 2(b^{(l+1)})^{\mathsf T} v + v^{\mathsf T} A^{(l+1)} v$ on $\mathbb{R}^{h_{l+1}}$. Completing the square at its minimiser $u^{(l+1)}$ gives $J_{l+1}(v) = \varepsilon_{l+1} + \|v - u^{(l+1)}\|^2_{A^{(l+1)}}$ for all v. Take $v = Pu^{(l)}$. By (12), $J_{l+1}(Pu^{(l)}) = 1 - 2(b^{(l)})^{\mathsf T} u^{(l)} + (u^{(l)})^{\mathsf T} A^{(l)} u^{(l)} = \varepsilon_l$, which is (24); the telescope follows by summation. For (25), (12) gives $V^{\mathsf T} r_l = b^{(l)} - A^{(l)} u^{(l)} = 0$, which is Galerkin orthogonality, and $A^{(l+1)}(u^{(l+1)} - Pu^{(l)}) = r_l$ turns (23) into (25). □

Two consequences of (25) deserve to be separated, because the whole difficulty of the subject sits in the difference between them.

Lemma 3.2 (the two dual descriptions of a rung)**.** *In the notation of Section 2.3, put*

$$s_l = W^{\mathsf T} b^{(l+1)}, \qquad \widehat{s}_l = s_l - B_x^{\mathsf T} u^{(l)} \;=\; W^{\mathsf T} r_l. \tag{26}$$

Then

$$\delta_l \;=\; \min_{x \in \mathbb{R}^{h_l}} \big(Vx + Wy^\star\big)^{\mathsf T} A^{(l+1)} \big(Vx + Wy^\star\big) \quad \textit{for a suitable } y^\star, \tag{27}$$

a minimum*, and*

$$\delta_l \;=\; \widehat{s}_l^{\mathsf T} S^{-1} \widehat{s}_l \;=\; \max_{v \ne 0} \frac{(\widehat{s}_l^{\mathsf T} v)^2}{v^{\mathsf T} S v}, \tag{28}$$

a maximum*.*

Proof. $\delta_l = \|u^{(l+1)} - Pu^{(l)}\|^2_{A^{(l+1)}}$ is the squared energy distance from $u^{(l+1)}$ to the coarse space, which is (27). For (28), (25) gives $r_l \in \operatorname{ran} W$, so $r_l = W\widehat{s}_l$ by (11), and $\delta_l = r_l^{\mathsf T}(A^{(l+1)})^{-1} r_l = \widehat{s}_l^{\mathsf T}(W^{\mathsf T}(A^{(l+1)})^{-1} W)\widehat{s}_l$. The block inverse formula for the splitting (17) gives $W^{\mathsf T}(A^{(l+1)})^{-1} W = S^{-1}$. The variational identity $\widehat{s}^{\mathsf T} S^{-1} \widehat{s} = \max_v (\widehat{s}^{\mathsf T} v)^2 / (v^{\mathsf T} S v)$ for $S \succ 0$ is Cauchy–Schwarz in the S-inner product, with equality at $v = S^{-1}\widehat{s}$. □

Remark 3.3 (the direction, stated pedantically)**.** From (27), every test vector yields an *upper* bound on δ_l; a minimum principle cannot supply a lower bound, no matter how good the test vector. From (28), every test vector yields a *lower* bound on δ_l, provided the denominator $v^{\mathsf{T}}Sv$ is replaced by something *larger*. This is the direction reversal of the title of Section 4: what a lower bound on δ_l needs is an *upper* bound on S, and upper bounds on Schur complements are cheap.

4 The certified rung

We need two classical upper-bound mechanisms.

Lemma 4.1 (Schur complement monotonicity)**.** *With the notation of* (17) *and* $A_c \succ 0$*, one has* $S \preceq A_z$*, and the only property of* A_c *used is* $A_c^{-1} \succeq 0$*, i.e. the* sign *of* A_c*.*

Proof. $A_z - S = B_x^{\mathsf{T}} A_c^{-1} B_x \succeq 0$ whenever $A_c^{-1} \succeq 0$. See [17]. □

Lemma 4.2 (Parseval majorant)**.** *Let* g_1, g_2 *be real, even and integrable on* $[-\pi, \pi]$ *with* $g_1 \le g_2$ *almost everywhere. Then* $T_M(g_1) \preceq T_M(g_2)$*, and consequently* $R^{\mathsf{T}} T_M(g_1) R \preceq R^{\mathsf{T}} T_M(g_2) R$ *for every real matrix* R *with* M *rows.*

Proof. For $v \in \mathbb{R}^M$ and $\mathcal{V}(\theta) = \sum_r v_r e^{ir\theta}$, Parseval gives $v^{\mathsf{T}} T_M(g) v = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(\theta) |\mathcal{V}(\theta)|^2 \, d\theta$. Apply this to $g = g_2 - g_1 \ge 0$. (That every nonnegative trigonometric polynomial is such a $|\mathcal{V}|^2$ is the Fejér–Riesz theorem [13, 26]; only the displayed identity is needed here.) □

Definition 4.3 (certified majorant)**.** Let $\sigma^{(M)}(\theta) = \sum_{|m| \le M-1} c_{|m|} e^{im\theta}$, so that $T_M(c) = T_M(\sigma^{(M)})$. A real even up $\in L^1[-\pi, \pi]$ is a *certified majorant* at level $l+1$ if up $\ge \sigma^{(M_{l+1})}$ pointwise. We then set

$$U_l \;=\; W^{\mathsf{T}} B^{\mathsf{T}} \, T_{M_{l+1}}(\mathrm{up}) \, B \, W. \tag{29}$$

Note that $A_z = W^{\mathsf{T}} B^{\mathsf{T}} T_{M_{l+1}}(c) B W$ by (5) and (17), so Lemma 4.2 applied with $R = BW$ gives $A_z \preceq U_l$ directly, and the Hankel-type term in (5) never has to be estimated: it is the reflection half of the isometry B.

Theorem 4.4 (the certified rung)**.** *Fix a level* l *and assume*

(H1) $A^{(l+1)} \succ 0$*;*

(H2) *the nesting* (12) *at level* l *— which by Lemma 2.7 holds for every consistent lag family, in particular for the class of Definition 2.8 — so that* $A_c = A^{(l)}$*;*

(H3) up *is a certified majorant at level* $l+1$ *and* $U_l \succ 0$*.*

Then, with $\widehat{s}_l$ *of* (26)*,*

$$\delta_l \;\ge\; \widehat{s}_l^{\mathsf{T}} U_l^{-1} \widehat{s}_l \;=:\; \widehat{c}_l \;\ge\; 0, \tag{30}$$

with equality in the second place only if $\widehat{s}_l = 0$*. The bound is attained by the explicit test vector* $v^\star = U_l^{-1} \widehat{s}_l$ *in* (28) *after replacing* S *by* U_l*. The proof uses no inverse, no smallest eigenvalue and no condition number of* A_c*, and no strengthened Cauchy–Schwarz constant.*

Proof. By (H1) and Section 2.3, $A_c \succ 0$ and $S \succ 0$, so Lemma 3.2 applies and $\delta_l = \widehat{s}_l^{\mathsf{T}} S^{-1} \widehat{s}_l$. By Lemma 4.1, $S \preceq A_z$; here (H2) is what makes $A_c \succeq 0$ the conclusion of the level below rather than a new hypothesis. By Definition 4.3 and Lemma 4.2, $A_z \preceq U_l$. Hence $0 \prec S \preceq U_l$, and Löwner's antitony of the matrix inverse on the positive definite cone [22] gives $U_l^{-1} \preceq S^{-1}$, i.e.

$$\delta_l = \widehat{s}_l^{\mathsf{T}} S^{-1} \widehat{s}_l \;\ge\; \widehat{s}_l^{\mathsf{T}} U_l^{-1} \widehat{s}_l = \widehat{c}_l.$$

Positivity of $\widehat{c}_l$ for $\widehat{s}_l \neq 0$ is $U_l \succ 0$. For the test vector, substituting $v = U_l^{-1} \widehat{s}_l$ into $(\widehat{s}_l^{\mathsf{T}} v)^2 / (v^{\mathsf{T}} U_l v)$ returns $\widehat{s}_l^{\mathsf{T}} U_l^{-1} \widehat{s}_l$ exactly, so the closed-form vector is optimal for the majorant. □

Remark 4.5 (what is certifiable, and how)**.** Each of (H1)–(H3) reduces to a completed Cholesky factorisation: $A^{(l+1)} \succ 0$ and $U_l \succ 0$ directly, and the majorant property through $T_{M_{l+1}}(\mathrm{up}) - T_{M_{l+1}}(c) \succeq 0$, which is Lemma 4.2 in the certified direction. In particular $A_c \succeq 0$ — the hypothesis whose quantitative version blocks the classical routes — is supplied by the previous rung's own conclusion and never has to be quantified. This is what makes a coarse-to-fine induction possible; see Corollary 5.6.

Remark 4.6 (the price of one test vector)**.** The bound (30) is the exact value of the maximum (28) with S replaced by the majorant U_l, so the entire loss relative to δ_l is the loss in $S \preceq U_l$ and none of it is a loss in the choice of test vector. The classical estimate of what a single test vector costs against the optimal one [20] is therefore quoted here only for the contrast: it does not enter.

Convention 4.7 (carried-solution accounting)**.** The numerator $\widehat{s}_l = s_l - B_x^{\mathsf{T}} u^{(l)}$ of (30) is a function of the coarse Galerkin solution $u^{(l)}$ carried along the ladder. Theorem 4.4 and Corollary 5.6 are therefore statements about the *carried-solution* rung: given the ladder and its coarse solutions, the bound is certified. A *solution-free* rung — one whose right-hand side is computable from (c, b, up) alone — is not claimed. See Proposition 6.1.

Remark 4.8 (the convention is a scope, not a circularity)**.** It is worth being explicit about what Convention 4.7 does and does not cost, because a bound whose right-hand side involves a solution invites the suspicion that it presupposes what it proves. It does not: $u^{(l)}$ is the Galerkin solution on the *coarse* space, computed there, and (30) bounds the increment gained by refining — so nothing in $\widehat{c}_l$ requires knowing ε_l, ε_{l+1}, or the fine solution $u^{(l+1)}$. This is precisely the information an a posteriori estimator has available: hierarchical estimators in the sense of [3, 5, 10] evaluate a residual against the computed coarse solution and are two-sided only under a saturation hypothesis, whereas Theorem 4.4 supplies the lower direction unconditionally, at the price of being a statement *given* the ladder. What the convention rules out is a different and stronger object — an a priori bound in the data alone — and Proposition 6.1 measures how far that object is from reach.

5 The composed chain

Theorem 4.4 bounds one rung; this section is its application. Two further ingredients turn the rungs into a statement that travels: the telescope of Lemma 3.1, which needs only semidefiniteness at the finest level, and a bordering step that grows the window. The composition below is conditional, and on the instance of Section 7 its hypotheses are certified window by window rather than proved for all windows; item (L2) of Section 8 says what would be needed for the latter.

5.1 Windows at one common resolution

Definition 5.1 (admissible common resolution)**.** Let $u_i < u_{i+1} < \cdots < u_j$ be consecutive comb nodes with log-gaps $g_k = u_{k+1} - u_k$, and fix $\nu \geq 4$. Put

$$D = \frac{\min_k g_k}{2\nu}, \qquad M_k = \min\{M \in 2\mathbb{N} : \ MD > u_k + D\}, \qquad \alpha_k = \frac{M_k D}{2}, \tag{31}$$

and let $A^{(k)}, b^{(k)}, \varepsilon_k, Q^{(k)}$ be the objects of Definition 2.1 at (D, M_k) with the lag sequence (20). We call D *admissible* for the run and write $g_c^{(k)} = (M_{k+1} - M_k)/2$ for the window increment per end.

The resolution is common to the whole run *by construction*, and this is the point: distinct nodes carry distinct natural resolutions $g_k/(2\nu)$, and those do not refine onto one another (a ratio $g_k/g_{k'}$ is a dyadic power only exceptionally — on the verified instance no consecutive gap ratio is dyadic, see Section 7). Choosing one D for the whole run buys an exactly nested family of windows at the price of the run length, since D is fixed by the *smallest* gap it contains.

Lemma 5.2 (the incoming node contributes exactly nothing)**.** *Let D be admissible and let u be a node with $u > 2\alpha_k = M_k D$ (i.e. a node not admitted at window k). Then*

$$S_D(mD, u) = 0 \qquad \text{for all } 0 \le m \le M_k - 1. \tag{32}$$

Consequently $c_m^{(k+1)} = c_m^{(k)}$ for all $0 \le m \le M_k - 1$.

Proof. $S_D(s,u) = \frac{1}{2}[\Delta_D(s-u) + \Delta_D(s+u)]$ is supported in $\{|s-u| < D\} \cup \{|s+u| < D\}$. For $s = mD \ge 0$ and $u > 0$ we have $s + u \ge u \ge \log 2 > D$ on the run (indeed $u \ge u_i$ and $D \le \log 2/(2\nu)$), so the second hat vanishes. For the first, $m \le M_k - 1$ gives $s \le (M_k - 1)D$, hence $u - s \ge M_k D - (M_k - 1)D = D$ and $\Delta_D(s-u) = 0$. The consequence is immediate from (20), because the two windows admit the same nodes with $u_j \le 2\alpha_k$, all newly admitted nodes satisfy $u > M_k D$, and $a(mD)$ does not depend on M. □

Lemma 5.3 (exact handover identity)**.** *Let $D > 0$, let $M_o < M_n$ be even with $M_n - M_o = 2g_c$, and let $c^{(o)}, c^{(n)}$ be lag sequences with $c_m^{(n)} = c_m^{(o)}$ for $0 \le m \le M_o - 1$. Let $A^{(o)}, b^{(o)}$ and $A^{(n)}, b^{(n)}$ be as in Definition 2.1 at (D, M_o) and (D, M_n), and write $h_o = M_o/2$. Then, with indices $r = g_c + r'$, $s = g_c + s'$ for $r', s' \in \{0, \dots, h_o - 1\}$,*

$$A_{rs}^{(n)} = A_{r's'}^{(o)}, \qquad b_r^{(n)} = b_{r'}^{(o)}. \tag{33}$$

Hence the trailing $h_o \times h_o$ block of $Q^{(n)} = A^{(n)} - b^{(n)}(b^{(n)})^{\mathsf{T}}$ equals $Q^{(o)}$ exactly.

Proof. For the form, $A_{rs}^{(n)} = c_{|r-s|}^{(n)} - c_{M_n-1-r-s}^{(n)}$ by (5). Now $|r-s| = |r'-s'| \le h_o - 1 \le M_o - 1$ and $M_n - 1 - r - s = M_n - 2g_c - 1 - r' - s' = M_o - 1 - r' - s'$, which lies in $\{1, \dots, M_o - 1\}$ because $r' + s' \le 2h_o - 2 = M_o - 2$. Both indices are therefore $\le M_o - 1$, so the hypothesis on the lag sequences applies and the right-hand side is $c_{|r'-s'|}^{(o)} - c_{M_o-1-r'-s'}^{(o)} = A_{r's'}^{(o)}$. For the pole vector, $\alpha_n - g_c D = (\frac{M_n}{2} - g_c)D = \frac{M_o}{2}D = \alpha_o$, so by (3) $\bar{x}_r^{(n)} = -\alpha_n + (g_c + r' + \frac{1}{2})D = -\alpha_o + (r' + \frac{1}{2})D = \bar{x}_{r'}^{(o)}$, and the prefactor $8D^{-1/2}\sinh(D/4)$ in (6) is unchanged because D is. The statement about Q follows. □

Remark 5.4. The two lemmas together say that on an admissible run the chain *composes literally*: by Lemma 5.2 the lag sequences agree on the old range, so by Lemma 5.3 the trailing block of the next window's bordered form is the current window's bordered form, not merely close to it. The coordinate ordering matters and is the natural one: index $r = 0$ is the outermost cell, so a window that grows by g_c cells per end acquires its new coordinates at the *leading* indices.

5.2 The margin-free bordering step

Lemma 5.5 (Albert's criterion)**.** *Let $\mathcal{M} = \left[\begin{smallmatrix} A & C \\ C^{\mathsf{T}} & X \end{smallmatrix}\right]$ be symmetric. Then $\mathcal{M} \succeq 0$ if and only if*

$$X \succeq 0, \qquad \operatorname{ran}(C^{\mathsf{T}}) \subseteq \operatorname{ran}(X), \qquad A - CX^{+}C^{\mathsf{T}} \succeq 0. \tag{34}$$

If $X \succ 0$ the middle condition is vacuous and $X^{+} = X^{-1}$.

Proof. [1]; the range condition is Douglas' lemma [11]. □

The point of (34) for us is negative: *nothing in it refers to the size of* $\lambda_{\min}(X)$. Contrast the norm-perturbation surrogate, which writes $\mathcal{M}$ as a perturbation of $\operatorname{diag}(A, X)$ and asks $\lambda_{\min}(X) > \|C\|_2^2/\lambda_{\min}(A)$ or the symmetric variant; that estimate divides by $\lambda_{\min}(X)$ and is therefore hostage to exactly the quantity a lower bound on ε is trying to produce. Section 7 reports that on the verified instance the surrogate is negative on every step while the exact Schur complement of (34) is of order 10^{-1}.

5.3 The chain

Corollary 5.6 (margin-free chain composition)**.** *Let D be admissible for the consecutive nodes $u_i, \dots, u_j$ in the sense of Definition 5.1. For each k let $D = D_0 > \cdots > D_{L_k}$ be the level ladder (8) at fixed α_k, and assume*

(A1) (base) $\varepsilon_{L_k}^{(k)} \geq 0$ *at the finest level of the chain of window k — equivalently, by (2), $Q^{(k,L_k)} \succeq 0$. This is pure semidefiniteness: no rate, no margin.*

(A2) *the hypotheses* (H1)–(H3) *of Theorem 4.4 at every rung of every chain, together with Convention 4.7;*

(A3) *for every k, the $g_c^{(k)} \times g_c^{(k)}$ Schur complement $A_{\mathrm{new}}^{(k+1)} - C_k\,(Q^{(k)})^{-1} C_k^{\mathsf{T}}$ is positive semidefinite, where $A_{\mathrm{new}}^{(k+1)}$ and C_k are the leading block and the coupling block of $Q^{(k+1)}$ in the splitting of Lemma 5.3.*

Then for every $k \in \{i, \dots, j\}$

$$\varepsilon_0^{(k)} \;\geq\; \varepsilon_{L_k}^{(k)} + \sum_{l<L_k} \widehat{c}_l^{(k)} \;\geq\; \sum_{l<L_k} \widehat{c}_l^{(k)} \;>\; 0, \qquad \textit{hence} \qquad Q^{(k)} \succ 0, \tag{35}$$

and the run is traversed

$$Q^{(i)} \succ 0 \implies Q^{(i+1)} \succ 0 \implies \dots \implies Q^{(j)} \succ 0. \tag{36}$$

Proof. Fix k. By (A2) and Theorem 4.4, $\delta_l^{(k)} \geq \widehat{c}_l^{(k)} > 0$ for every rung; by Lemma 3.1, $\varepsilon_0^{(k)} = \varepsilon_{L_k}^{(k)} + \sum_l \delta_l^{(k)}$; and by (A1), $\varepsilon_{L_k}^{(k)} \geq 0$. This is (35), and $Q^{(k)} \succ 0$ follows from (2) with strict inequality. For (36), Lemmas 5.2 and 5.3 put $Q^{(k+1)}$ in the block shape $\left[\begin{smallmatrix} A_{\mathrm{new}}^{(k+1)} & C_k \\ C_k^{\mathsf{T}} & Q^{(k)} \end{smallmatrix}\right]$ with the trailing block *equal* to $Q^{(k)}$. Apply Lemma 5.5 with $X = Q^{(k)}$: the first condition is the previous step's conclusion, the second is vacuous because $Q^{(k)} \succ 0$, and the third is (A3). □

Remark 5.7 (where the induction runs, and why it must)**.** The telescope (24) places its certified bound on the *coarse* level: $\varepsilon_{\mathrm{coarse}} = \varepsilon_{\mathrm{fine}} + \delta$ with $\delta \geq 0$, so the target window must be the coarsest level of its chain and the base case (A1) sits at the finest. A bound at the fine level would require certified *upper* bounds on the rungs, i.e. a strengthened Cauchy–Schwarz constant, which is precisely the quantity the construction is built to avoid. This is a structural feature, not a choice.

6 A measured obstruction: the solution-free rung

Convention 4.7 is the one hypothesis of Corollary 5.6 that is neither an identity nor a Cholesky certificate, so it deserves to be examined rather than declared. The natural attempt to remove it is to replace the optimal test vector $v^\star = U^{-1}\widehat{s}$, which needs $u^{(l)}$, by the fully explicit $w = U^{-1}s$, which does not.

Proposition 6.1 (the solution-free rung, and how it fails)**.** *With $w = U_l^{-1} s_l$, the test vector w in* (28) *gives the valid lower bound*

$$\delta_l \;\geq\; \widehat{c}_l^{\,\mathrm{expl}} \;=\; \frac{(\widehat{s}_l^{\mathsf{T}} w)^2}{s_l^{\mathsf{T}} w} \;=\; (1-\rho_l)^2\,(s_l^{\mathsf{T}} U_l^{-1} s_l), \qquad \rho_l = \frac{(u^{(l)})^{\mathsf{T}} B_x w}{s_l^{\mathsf{T}} U_l^{-1} s_l}, \tag{37}$$

in which the factor $s_l^{\mathsf{T}} U_l^{-1} s_l$ is computable from (c, b, up) alone and the coarse solution enters through the single scalar ρ_l. Consequently the rung is solution-free up to a constant if and only if *ρ_l admits a bound away from* 1*.*

On the verified instance of Section 7 it does not, and the failure is by cancellation rather than by a small margin:

(i) *$|\rho_l|$ is measured in* $[0.3888, 14.8018]$ *over the* 88 *rungs of the assembled ladder (*$[0.88, 2.61]$ *on the separate level chain of the same family), so ρ_l is not a perturbation of* 0*;*

(ii) *$\|\widehat{s}_l\|^2/\|s_l\|^2$ is measured in $[6.2\cdot10^{-1},\, 1.8\cdot10^{5}]$ — below* 1 *at small α and above it at large α — so the coupling term does not act as a fraction of the explicit high-pass part at all;*

(iii) *the norm-only variant, which replaces $\widehat{s}_l$ by $\|s_l\| - \|B_x^{\mathsf{T}} u^{(l)}\|$ under Cauchy–Schwarz, returns exactly* 0.0 *of δ_l on every one of the* 88 *rungs.*

Proof of the algebraic part. w is a test vector in (28) after replacing S by U_l, which is legitimate by the chain $0 \prec S \preceq U_l$ established in the proof of Theorem 4.4; hence $\delta_l \geq (\widehat{s}_l^{\mathsf{T}} w)^2/(w^{\mathsf{T}} U_l w)$ and $w^{\mathsf{T}} U_l w = s_l^{\mathsf{T}} U_l^{-1} s_l$. Substituting $\widehat{s}_l = s_l - B_x^{\mathsf{T}} u^{(l)}$ gives $\widehat{s}_l^{\mathsf{T}} w = s_l^{\mathsf{T}} U_l^{-1} s_l - (u^{(l)})^{\mathsf{T}} B_x w = (1-\rho_l)\, s_l^{\mathsf{T}} U_l^{-1} s_l$, and (37) follows. □

The items (i)–(iii) are measurements on a finite ladder and are reported as such; they do not prove that no solution-free rung exists. What they do establish is the *shape* of the obstruction, and that shape is informative: since ρ_l is of order one and larger, the coupling term $B_x^{\mathsf{T}} u^{(l)}$ cancels the explicit high-pass part s_l rather than perturbing it. A solution-free rung therefore needs a bound on the *direction* of $\widehat{s}_l$, not on its size, and any estimate that proceeds through norms is provably worthless here — item (iii) is that statement, measured.

7 Numerical verification

7.1 Floating-point convention

Every positivity statement below is a completed Cholesky factorisation relative to a declared floor. With $u = 2^{-53} = 1.110\cdot10^{-16}$ the unit roundoff and $c_h = (h+1)/(1-(h+1)u)$, a completed Cholesky factorisation of $A - \delta I$ with $\delta = c_h u \|A\|_\infty$ certifies $\lambda_{\min}(A) > 0$ [31, 18]; the cheap certified bound $\|A\|_2 \leq \|A\|_\infty = \max_r \sum_s |A_{rs}|$ is used for the norm [15]. At the largest size occurring below, $h = 1495$, the floor is $1.66\cdot10^{-13}\,\|A\|_\infty$. Quantities obtained from a symmetric eigenvalue routine are labelled *measured* and are never used as certificates; every reported *head-room* is a factor over the declared floor, so any value above 1 certifies the corresponding sign.

7.2 The verified instance

Table 1 lists the instance. Table 2 lists what was certified on it. We highlight five entries.

Identities. The nesting (12), proved in Lemmas 2.7 and 2.9, is reproduced to relative $3.0\cdot10^{-14}$ for the form and $6.2\cdot10^{-16}$ for the pole vector on all 88 rungs; since it is an identity, these residuals compare two orders of summation for the same exact quantity and are a check on the assembly code, not on the statement. The saturation identity, the dual-residual identity (25) and Galerkin orthogonality hold to $4.7\cdot10^{-7}$; the telescope $\sum_l \delta_l = \varepsilon_0 - \varepsilon_L$ to $3.3\cdot10^{-7}$;

quantity	value
common resolution	$D = 3.472446 \cdot 10^{-3}$, from $\min_k g_k = 0.027780$ and $\nu = 4$; effective $\nu_{\text{eff}} = g_k/(2D) = 4.0 \ldots 58.4$
run of nodes	30 consecutive prime-power nodes, $n = 2, \ldots, 79$; $\alpha_k = 0.3507 \ldots 2.1494$; $h_k = 101 \ldots 619$; $g_c^{(k)} = 4 \ldots 58$ cells per end
gap facts used	$g_k \le \log 2$ (max. observed 0.405465) and $g_k \ge \log(1 + 1/n_k)$ (max. observed $1/g_k = 1.4985 \cdot 10^5$)
window nesting	$M_{k+1}^{(\text{old})} = M_k^{(\text{new})}$ on all 29 consecutive pairs, an integer identity
non-dyadic gaps	0 of 84 consecutive gap ratios g_k/g_{k+1} is a dyadic power: per-node resolutions have no common refinement
ladder size	52 windows (30 composed + 22 at per-node resolutions, 51 distinct (n, D) pairs), 88 rungs, chains of 2–5 levels, plus 10 bordering-only windows up to $n = 1331$
largest factorisation	$h = 1495$ (cap 1500); finest chain level $h \le 1400$; base-case sizes $h = 416 \ldots 1384$
certificates	430 completed Cholesky factorisations, 440 verified identities, all completing
runtime	8.2 s

Table 1: The verified instance. All entries are outputs of the single script named in Section 7.5.

the closed-form piecewise-constant lag formula to $2.2 \cdot 10^{-14}$. The bit-exact part of Lemma 5.2 is bit-exact in the run: the incoming node's contribution restricted to the old lag range has maximum absolute entry $0.0 \cdot 10^0$, and the trailing block of $Q^{(k+1)}$ agrees with $Q^{(k)}$ to relative $2.9 \cdot 10^{-14}$, which is the roundoff of two different summation orders for the same exact quantity (Lemma 5.3).

The rung. All three certificates of Theorem 4.4 complete on 88 of 88 rungs, and $0 < \widehat{c}_l \le \delta_l$ throughout with retention $\widehat{c}_l/\delta_l \in [0.90709, 0.99708]$. The direction control of Remark 3.3 is also verified: the primal form (27), evaluated at the same test vector, returns values *above* δ_l on 88 of 88 rungs, i.e. it is an upper bound and can never serve as a supply.

The lower bound. $\varepsilon_0^{(k)} \ge \sum_l \widehat{c}_l^{(k)} > 0$ on all 52 windows. Two normalisations are reported, and the second is the informative one. Against the measured ε_0 the retention is $0.0013 \ldots 0.9694$; but an L-level chain cannot supply more than it telescopes, so the relevant ceiling is $1 - \varepsilon_L/\varepsilon_0$, measured at $0.001295 \ldots 0.993502$, and against that ceiling the certified fraction is $0.9140 \ldots 0.9929$ on every window. The certified margin exceeds the *declared* floating-point bound $c_h u \,\mathrm{cond}(A_0^{(k)})$ on ε by a factor $31.6 \ldots 8.7 \cdot 10^{10}$ at $\mathrm{cond}(A_0) = 6.9 \cdot 10^1 \ldots 3.8 \cdot 10^5$. The thinnest window of the ladder is $n = 243$ ($\alpha = 2.7492$, $h_0 = 679$) at 31.6, and it is thin because its retention 0.6054 is pinned to the small ceiling 0.629042 there, not because the arithmetic is marginal.

The base case. At the finest level of each chain, completed Cholesky factorisations of $A^{(L)}$ and of $Q^{(L)} = A^{(L)} - b^{(L)}(b^{(L)})^{\mathsf{T}}$ both run, at sizes $h = 416 \ldots 1384$, with $\lambda_{\min}(Q^{(L)})$ at $4.4 \cdot 10^5 \ldots 5.6 \cdot 10^{10}$ times its own floor; and $\operatorname{sign} \lambda_{\min}(Q^{(L)}) = \operatorname{sign} \varepsilon_L$ on 52 of 52 windows, which is (2) verified rather than assumed.

The bordering step. On 62 of 62 steps ($n = 2, \ldots, 1331$, $h_{\text{new}} = 30 \ldots 1495$) the trailing block is positive definite, the range condition of Lemma 5.5 is vacuous, and the exact $g_c \times g_c$ Schur complement has $\lambda_{\min} \in [1.4839 \cdot 10^{-3}, 1.2528 \cdot 10^{-1}]$, a factor $3.0 \cdot 10^{10} \ldots 4.6 \cdot 10^{13}$ above its own floor. The norm-perturbation surrogate of the *same* Schur complement, which divides by $\lambda_{\min}$ of the trailing block, is negative on all 62 steps ($-1.04 \cdot 10^7 \cdots - 1.29 \cdot 10^1$). Lemma 5.5 never performs that division.

statement	certified / measured on the instance
$P^{\mathsf{T}} A^{(l+1)} P = A^{(l)}$; $P^{\mathsf{T}} b^{(l+1)} = b^{(l)}$	identity (Lem. 2.7, 2.4, 2.9); reproduced to $3.0 \cdot 10^{-14}$ / $6.2 \cdot 10^{-16}$ on 88/88 rungs
$\varepsilon_0 = \varepsilon_L + \sum \delta_l$	identity; residual $3.3 \cdot 10^{-7}$
$\delta_l \geq \widehat{c}_l > 0$ (Thm. 4.4)	three completed Choleskys on 88/88 rungs; $\widehat{c}_l/\delta_l = 0.90709 \ldots 0.99708$
primal form is an upper bound (Rem. 3.3)	above δ_l on 88/88 rungs
$\varepsilon_0 \geq \sum \widehat{c}_l > 0$	52/52 windows; $0.9140 \ldots 0.9929$ of the L-level ceiling; head-room $31.6 \ldots 8.7 \cdot 10^{10}$ over the declared floor
base case $Q^{(L)} \succeq 0$	Cholesky at $h = 416 \ldots 1384$; head-room $4.4 \cdot 10^{5} \ldots 5.6 \cdot 10^{10}$
bordering step (Lem. 5.5)	62/62 steps; exact Schur complement $\lambda_{\min} = 1.48 \cdot 10^{-3} \ldots 1.25 \cdot 10^{-1}$, head-room $3.0 \cdot 10^{10} \ldots 4.6 \cdot 10^{13}$
norm-perturbation surrogate	negative on 62/62 steps
solution-free rung (Prop. 6.1)	$\lvert\rho_l\rvert = 0.3888 \ldots 14.8018$; norm-only bound returns 0.0 of the rung on 88/88

Table 2: What was certified, and what was measured. Head-room is always a factor over the declared floating-point floor of Section 7.

7.3 Two fitted trends, labelled as fits

The following are least-squares fits with jackknife bands. They are reported for orientation only and are *not* bounds; nothing in Theorem 4.4 or Corollary 5.6 depends on them.

$$\log(\widehat{c}_l/\delta_l) = -0.082 - 0.010 \log D + 0.001 \log \alpha, \quad \text{rms } 0.012, \ \text{bands } \pm 0.004, \ \pm 0.005; \tag{38}$$

$$\log\Big(\textstyle\sum_l \widehat{c}_l / \sum_l \delta_l\Big) = -0.018 - 0.012 \log \alpha, \qquad \text{rms } 0.004, \ \text{band } \pm 0.002. \tag{39}$$

Equation (39) is fitted on the composed run only, where D is common by construction so no exponent in D is identifiable. Read together, they say that on this ladder the certified fraction of the telescope does not thin out along the run.

7.4 The near-null direction

The measurement that motivates Section 1's claim about symbol methods is the following, taken on the same family of forms at the two-level stage. Let ℓ be a certified pointwise minorant of the symbol, $\ell \leq \sigma^{(M)}$, so that $T_M(\ell) \preceq T_M(c)$ by Lemma 4.2, and let $E_c = V^{\mathsf{T}} B^{\mathsf{T}} T_M(\ell) B V$ be the coarse block it produces. Then $\lambda_{\min}(E_c) < 0$ on every tested row, so the minorant certifies nothing about A_c; whereas $\lambda_{\min}(A_c)$ itself is positive, measured in $[1.7 \cdot 10^{-5}, 2.9 \cdot 10^{-4}]$, at condition numbers up to $3.1 \cdot 10^{7}$ — two to five orders of magnitude below the size the minorant would have to reproduce. The near-null eigenvector of A_c is smooth and lies in $\operatorname{ran} V$: its small Rayleigh quotient is produced by cancellation between the Toeplitz entries, not by a small value of the symbol along it, and a pointwise minorant is by construction blind to that. This is exactly the quantity that Theorem 4.4 does not need, since it uses A_c only through $A_c \succeq 0$.

7.5 Reproducibility

All numbers in Tables 1 and 2, and item (iii) and the first band of item (i) of Proposition 6.1, are produced by one self-contained Python script (`grand_assembly_probe.py`, supplied as supplementary material) using only `numpy` and `scipy` (`cho_factor`, `eigvalsh`). It performs no file writes, runs in about 8.2 s on a current laptop, and reports 34 internal checks, all passing.

Every matrix that is factorised, inverted or diagonalised has $h \leq 1500$; the only larger objects are matrix-free fast Fourier transforms used to evaluate symbols, at grid size up to 2^{20}. The two remaining sets of numbers — the second band of Proposition 6.1(i), measured on a single level chain rather than on the assembled ladder, and the two-level measurements of Section 7.4 — come from two further scripts of the same family, supplied alongside it; they are reported separately for exactly that reason.

8 Limitations, and what is not claimed

(L1) **The run is finite, and its length is structurally bounded.** By Definition 5.1 the common resolution is $D = \min_k g_k/(2\nu)$, the window sizes are $M_k \approx u_k/D$, and the cost of a factorisation is cubic in M_k. A longer run therefore forces a smaller D and larger windows, and the run verified here — 30 consecutive nodes, $n = 2, \ldots, 79$ — is what fits under a matrix cap of 1500. Nothing is claimed about nodes outside the tested set: of the 34 034 comb nodes tabulated up to $n = 400\,000$, exactly 48 were touched at all, and 62 of the 85 admissible bordering steps in that table were certified.

(L2) **No uniformity in the window index.** This is the main gap, and it is a gap in *uniformity*, not in size. Specifically: hypothesis (A3) of Corollary 5.6 is verified window by window and we have no uniform-in-k lower bound for the $g_c \times g_c$ Schur complement; and we have no uniform lower bound for $\widehat{c}_l$. Without both, (36) does not extend to an infinite family, and we make no such claim. Consequently Corollary 5.6 should be read as we have labelled it, an application of Theorem 4.4 together with a certified instance of its hypotheses, and not as a theorem about the family of all windows. Theorem 4.4 itself is unaffected: it is a statement about one rung, with hypotheses that are checkable on that rung.

(L3) **The base case is a certificate, not a theorem.** Hypothesis (A1) is a completed Cholesky factorisation of one explicit matrix of size $h = 416 \ldots 1384$ per window. It is verified, not proved.

(L4) **Convention 4.7 stands.** The certified rung is the carried-solution rung. Proposition 6.1 shows that removing the convention requires a bound on the direction of $\widehat{s}_l$, and that norm-based attempts fail by cancellation on the tested ladder; it does not show that no such bound exists.

(L5) **A second, independent route is deliberately excluded.** The increment δ_l also admits a lower bound through a corner estimate for the oscillation of the Galerkin solution — a one-cell Poincaré step in the spirit of [25] followed by a Cauchy–Schwarz reduction to the corner mass — which requires a sign hypothesis on the increment sequence and a summed Harnack-type ratio bound. Both hold on the windows we measured and neither is proved uniformly; their classical address is the corner regularity of inverses of Toeplitz forms with singular symbols [30, 14, 28] for the ratio, and the discrete maximum principle for monotone matrices [24, 29] for the sign. Since that route is logically independent of Theorem 4.4 and Corollary 5.6, which do not use it, we omit it here and defer it to a separate note whose subject is a discrete Harnack inequality for the increments of the inverse.

(L6) **Fits are fits.** Equations (38) and (39), and every exponent quoted for orientation, are least-squares fits with jackknife bands, not bounds.

(L7) **The example class is an example.** Theorem 4.4 and Corollary 5.6 are symbol-agnostic; the class of Definition 2.8 enters only through the verification, and the arithmetic in it enters only as the source of the comb positions and through the two classical gap

inequalities named in Section 2.4(iii). No asymptotic property of the symbol family and no arithmetic statement is used or claimed.

Acknowledgements

The construction grew out of a long sequence of failed attempts, and the two that were most instructive are recorded above: the primal form of a rung (Remark 3.3, an upper bound where a lower bound was wanted) and the norm-perturbation surrogate of the bordering step (Section 5, negative on every step). Both are reported as measurements because both are, in retrospect, the reason the working argument has the shape it has.

References

[1] A. Albert, *Conditions for positive and nonnegative definiteness in terms of pseudoinverses*, SIAM J. Appl. Math. **17** (1969), 434–440.

[2] O. Axelsson and P. S. Vassilevski, *Algebraic multilevel preconditioning methods, I*, Numer. Math. **56** (1989), 157–177.

[3] R. E. Bank and R. K. Smith, *A posteriori error estimates based on hierarchical bases*, SIAM J. Numer. Anal. **30** (1993), 921–935.

[4] A. Böttcher and B. Silbermann, *Analysis of Toeplitz Operators*, 2nd ed., Springer, Berlin, 2006.

[5] F. A. Bornemann, B. Erdmann and R. Kornhuber, *A posteriori error estimates for elliptic problems in two and three space dimensions*, SIAM J. Numer. Anal. **33** (1996), 1188–1204.

[6] J. H. Bramble, J. E. Pasciak and J. Xu, *Parallel multilevel preconditioners*, Math. Comp. **55** (1990), 1–22.

[7] A. Brandt, *Multi-level adaptive solutions to boundary-value problems*, Math. Comp. **31** (1977), 333–390.

[8] J. Céa, *Approximation variationnelle des problèmes aux limites*, Ann. Inst. Fourier **14** (1964), 345–444.

[9] P. L. Chebyshev, *Mémoire sur les nombres premiers*, J. Math. Pures Appl. **17** (1852), 366–390.

[10] W. Dörfler and R. H. Nochetto, *Small data oscillation implies the saturation assumption*, Numer. Math. **91** (2002), 1–12.

[11] R. G. Douglas, *On majorization, factorization, and range inclusion of operators on Hilbert space*, Proc. Amer. Math. Soc. **17** (1966), 413–415.

[12] J. Durbin, *The fitting of time series models*, Rev. Inst. Int. Statist. **28** (1960), 233–244.

[13] L. Fejér, *Über trigonometrische Polynome*, J. Reine Angew. Math. **146** (1916), 53–82.

[14] M. E. Fisher and R. E. Hartwig, *Toeplitz determinants: some applications, theorems, and conjectures*, Adv. Chem. Phys. **15** (1969), 333–353.

[15] S. Gershgorin, *Über die Abgrenzung der Eigenwerte einer Matrix*, Izv. Akad. Nauk SSSR Ser. Mat. **6** (1931), 749–754.

[16] U. Grenander and G. Szegő, *Toeplitz Forms and Their Applications*, University of California Press, Berkeley, 1958.

[17] E. V. Haynsworth, *Determination of the inertia of a partitioned Hermitian matrix*, Linear Algebra Appl. **1** (1968), 73–81.

[18] N. J. Higham, *Accuracy and Stability of Numerical Algorithms*, 2nd ed., SIAM, Philadelphia, 2002.

[19] M. Kac, W. L. Murdock and G. Szegő, *On the eigenvalues of certain Hermitian forms*, J. Rational Mech. Anal. **2** (1953), 767–800.

[20] L. V. Kantorovich, *Functional analysis and applied mathematics*, Uspekhi Mat. Nauk **3** (1948), 89–185.

[21] N. Levinson, *The Wiener RMS error criterion in filter design and prediction*, J. Math. Phys. **25** (1947), 261–278.

[22] K. Löwner, *Über monotone Matrixfunktionen*, Math. Z. **38** (1934), 177–216.

[23] H. L. Montgomery and R. C. Vaughan, *Hilbert's inequality*, J. London Math. Soc. (2) **8** (1974), 73–82.

[24] A. M. Ostrowski, *Über die Determinanten mit überwiegender Hauptdiagonale*, Comment. Math. Helv. **10** (1937), 69–96.

[25] L. E. Payne and H. F. Weinberger, *An optimal Poincaré inequality for convex domains*, Arch. Rational Mech. Anal. **5** (1960), 286–292.

[26] F. Riesz, *Über ein Problem des Herrn Carathéodory*, J. Reine Angew. Math. **146** (1916), 83–87.

[27] G. Szegő, *Orthogonal Polynomials*, Amer. Math. Soc. Colloquium Publications **23**, New York, 1939.

[28] W. F. Trench, *An algorithm for the inversion of finite Toeplitz matrices*, J. Soc. Indust. Appl. Math. **12** (1964), 515–522.

[29] R. S. Varga, *Matrix Iterative Analysis*, Prentice-Hall, Englewood Cliffs, 1962.

[30] H. Widom, *Asymptotic behavior of block Toeplitz matrices and determinants*, Adv. Math. **13** (1974), 284–322.

[31] J. H. Wilkinson, *Rounding Errors in Algebraic Processes*, Prentice-Hall, Englewood Cliffs, 1963.

[32] H. Yserentant, *On the multi-level splitting of finite element spaces*, Numer. Math. **49** (1986), 379–412.